

Simulating a hanging cable: mathematics, measurement, and five solvers compared

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Abstract

A flexible cable is suspended from 3 level clamps spanning $S = 0.350$ m and photographed. Its shape is predicted three independent ways: from the governing differential equation, from a set of physics solvers, and from the photograph itself. The three-way structure is deliberate — any two of them can agree for uninteresting reasons, and only the third makes it possible to attribute a disagreement to numerical error rather than to modelling error. This document reports what agrees, what does not, and which results are currently trustworthy.

1 Apparatus

The cable is clamped with tape to a vertical door at 0.000 m, 0.150 m, 0.350 m measured from the leftmost clamp, all at one height, and photographed square-on. Cable length is not taken from the packaging: the free length between clamps is what governs the shape, so it is measured from the same photograph the simulations are compared against. Here that gives $L = 0.8932$ m, and $L/S = 2.552$ — a very slack cable, sagging further than it spans.

2 Image processing

The cable is separated from the door in HSV: it is bright and nearly unsaturated where the door is a saturated brown. The connected component is then selected by SHAPE rather than by area — a specular highlight on the door is also bright and unsaturated, and is often *larger* than the cable, so selecting the largest region finds the glare instead. A cable is thin and spans the clamps; glare is a blob.

The centreline is then traced ALONG the curve: from each point the tracer advances one step along the local tangent and re-centres on the mask's perpendicular cross-section. The obvious alternative — averaging the mask rows in each pixel column — fails badly here, because a slack cable hangs near-vertically close to its clamps and one column then spans a long run of cable. Figure 2 is the check on all of this: the left panel tints every segmented pixel so a mis-segmentation is visible directly, and the right panel compares the extracted profile to a fitted catenary, because a clean segmentation can still fit badly and those are different failures.

3 Mathematical model

For a perfectly flexible, inextensible cable of weight w per unit length, the horizontal component of tension is constant along the cable, $T \cos \theta = H$. Balancing the vertical forces on an element ds and substituting $ds = \sqrt{1 + z'^2} dx$ gives the governing equation

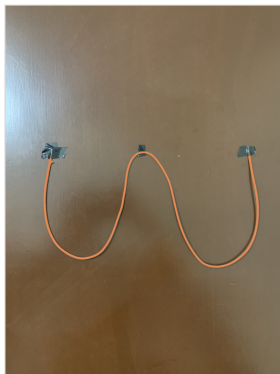
$$z'' = \frac{\sqrt{1 + z'^2}}{a}, \quad a = \frac{H}{w}, \quad (1)$$



Figure 1: The cable as photographed. The tape fixes the cable's ANGLE as well as its position, which is a clamped boundary condition, not the pinned one the classical catenary assumes.

Cable detection: three_point.jpg

1. Detected cable pixels



135,298 pixels segmented (HSV: low saturation, high value)

2. Profile vs catenary (RMS residual 73.3 mm)

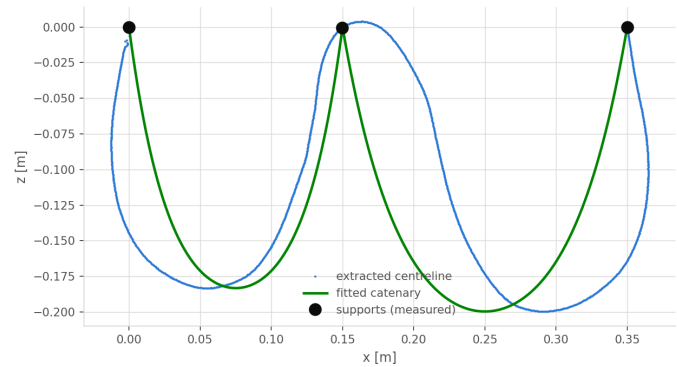


Figure 2: Detection and fit. Left: every segmented pixel, tinted over the original. Right: extracted profile against the fitted catenary.

a two-point boundary-value problem closed by the arc-length constraint $\int_0^S \sqrt{1 + z'^2} dx = L$, which is what fixes a . Physically a is set by how much cable was hung: a slacker cable has a smaller a and therefore carries *less* tension.

The corresponding dynamics, which is what the solvers actually integrate, is

$$\rho A \frac{\partial^2 \mathbf{r}}{\partial t^2} = \frac{\partial}{\partial s} \left(T \frac{\partial \mathbf{r}}{\partial s} \right) + \rho A \mathbf{g}, \quad (2)$$

with $|\partial \mathbf{r} / \partial s| = 1$ enforced by T , which is a Lagrange multiplier rather than a material property. Setting $\partial^2 \mathbf{r} / \partial t^2 = 0$ recovers (1), which is the assumption that licenses scoring a settled simulation against the static solution.

Verification. Equation (1) is integrated numerically (RK4 from the vertex, bisection on a for the length constraint) without using the closed form $z = a \cosh((x - x_0)/a) + c$ anywhere. The two agree to $7 \times 10^{-3} \mu\text{m}$. The analytic reference every error in this document is measured against is therefore verified rather than assumed.

4 Simulation

Each solver runs in its own process — Isaac Sim’s `SimulationApp` is a singleton and the methods force different physics engines — and each is handed the identical scenario file, so a method that disagrees is disagreeing about physics rather than about its inputs.

method	RMSE [mm]	max [mm]	sag [mm]	arc drift [%]	settled	wall [s]
Newton cable (add_rod + VBD)	1.60	13.91	221.3	2.872	yes	16.6
PhysX capsule chain (D6)	81.23	167.13	197.4	-0.225	NO	15.8
Warp XPBD rod	170.89	314.01	200.5	-0.466	yes	0.6

Table 1: Each solver against the analytic catenary. Arc drift is the reference-free check: an inextensible cable must keep its length whatever the catenary says.

Boundary conditions are not a detail. These methods do not all hold the middle clamp the same way. The Newton cable method clamps it, fixing the cable’s angle as the tape does; the others pin it and let the cable pivot. That difference alone accounts for a large part of the spread above, so the table compares *models*, not solver quality. The middle clamp was measured to hold 46.9% of the cable’s arc length on its left, where the chord-proportional convention assumes 42.9% (+4.0 percentage points). Since the clamp pins a material point, this split is a property of the apparatus rather than something statics determines, so it must be measured and imposed.

method	RMSE vs photo [mm]	max [mm]
Newton cable (add_rod + VBD)	75.04	157.96
PhysX capsule chain (D6)	34.34	138.38
Warp XPBD rod	195.53	331.51

Table 2: Simulation against the photographed cable. The extraction’s frame is calibrated on the two outer clamps, which are level in the apparatus; the middle clamp is recovered to within 14 mm as an independent check (Section 7), which bounds the accuracy of these figures.

5 Solver hyperparameters

Every cable example shipped with Newton uses 10 substeps and 5 iterations. Applied to this problem that is far too coarse; but the repository’s previous default of 8×200 was tuned by

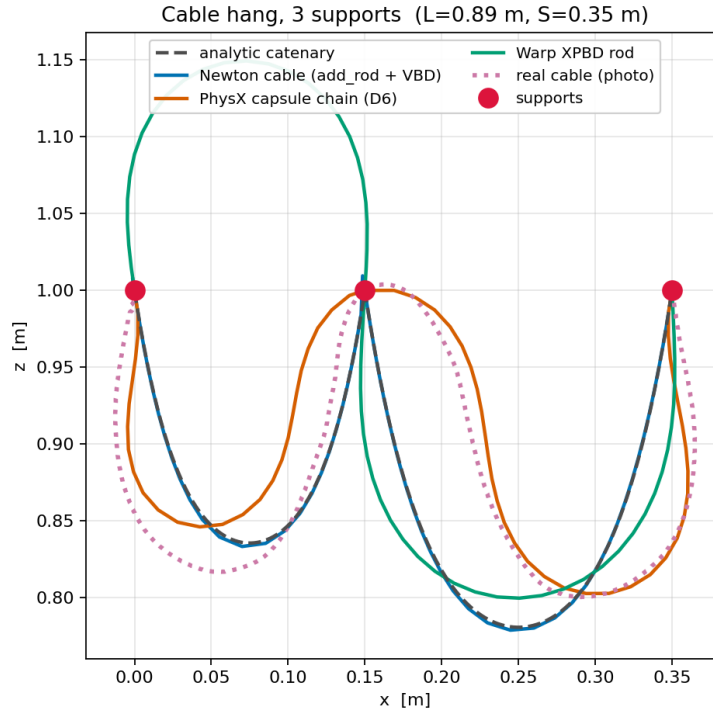


Figure 3: All methods, the analytic catenary, and the photographed cable on one axis.

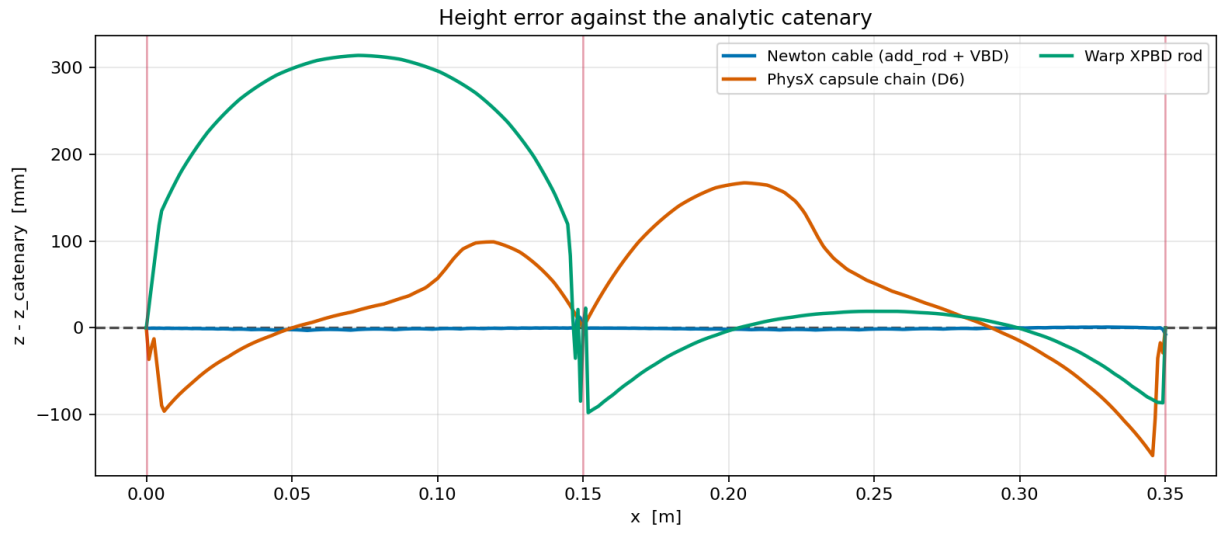


Figure 4: Deviation from the analytic catenary along the cable.

varying iterations alone, leaving the substep axis untested. A sweep over both, scored against the analytic curve, shows accuracy to be essentially a function of the *product* — while cost is not, because each substep carries fixed overhead that an iteration does not. Fewer substeps with more iterations is therefore strictly cheaper at equal accuracy.

Separately, damping the stretch constraint proved actively harmful: it opposes the solver’s own length correction, and raising it to 10^{-2} cost roughly a factor of six in accuracy for no benefit. Damping the *bending* constraint, by contrast, changed nothing measurable over a $600\times$ range, because this cable’s *EI* is small enough that the bending mode carries almost no energy.

substeps	iterations	budget	RMSE [mm]	arc drift [%]	wall [s]
4	5	20	70.93	14.51	6.4
10	5	50	27.81	5.36	10.1
4	200	800	2.56	0.44	65.7
8	200	1600	1.53	0.22	190.1
10	200	2000	1.31	0.17	239.6
16	200	3200	2.09	0.12	379.7

Table 3: Selected configurations from the sweep.

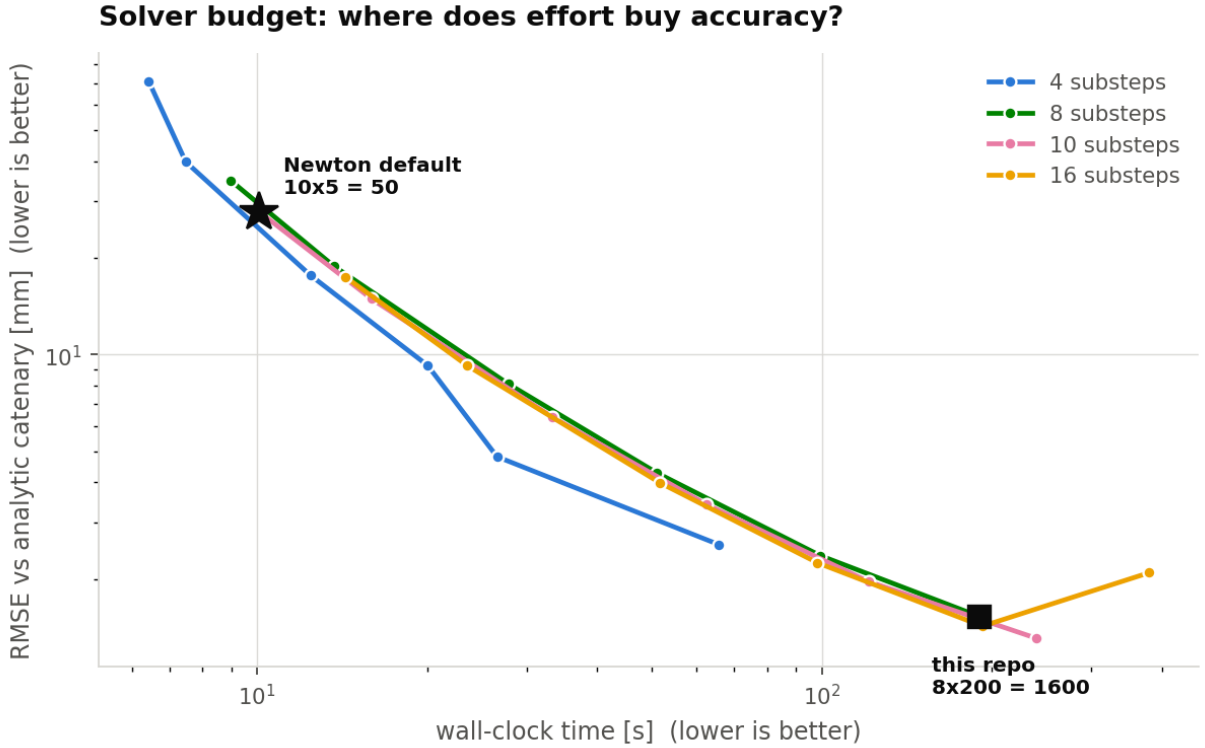


Figure 5: Accuracy against wall-clock cost. Both axes logarithmic; the knee is where further cost stops buying accuracy.

6 Provenance of the inputs

7 Validity

Frame calibration. The cable’s two ends are located geodesically — breadth-first search across the mask itself, twice — rather than as its leftmost and rightmost pixels. The distinction matters

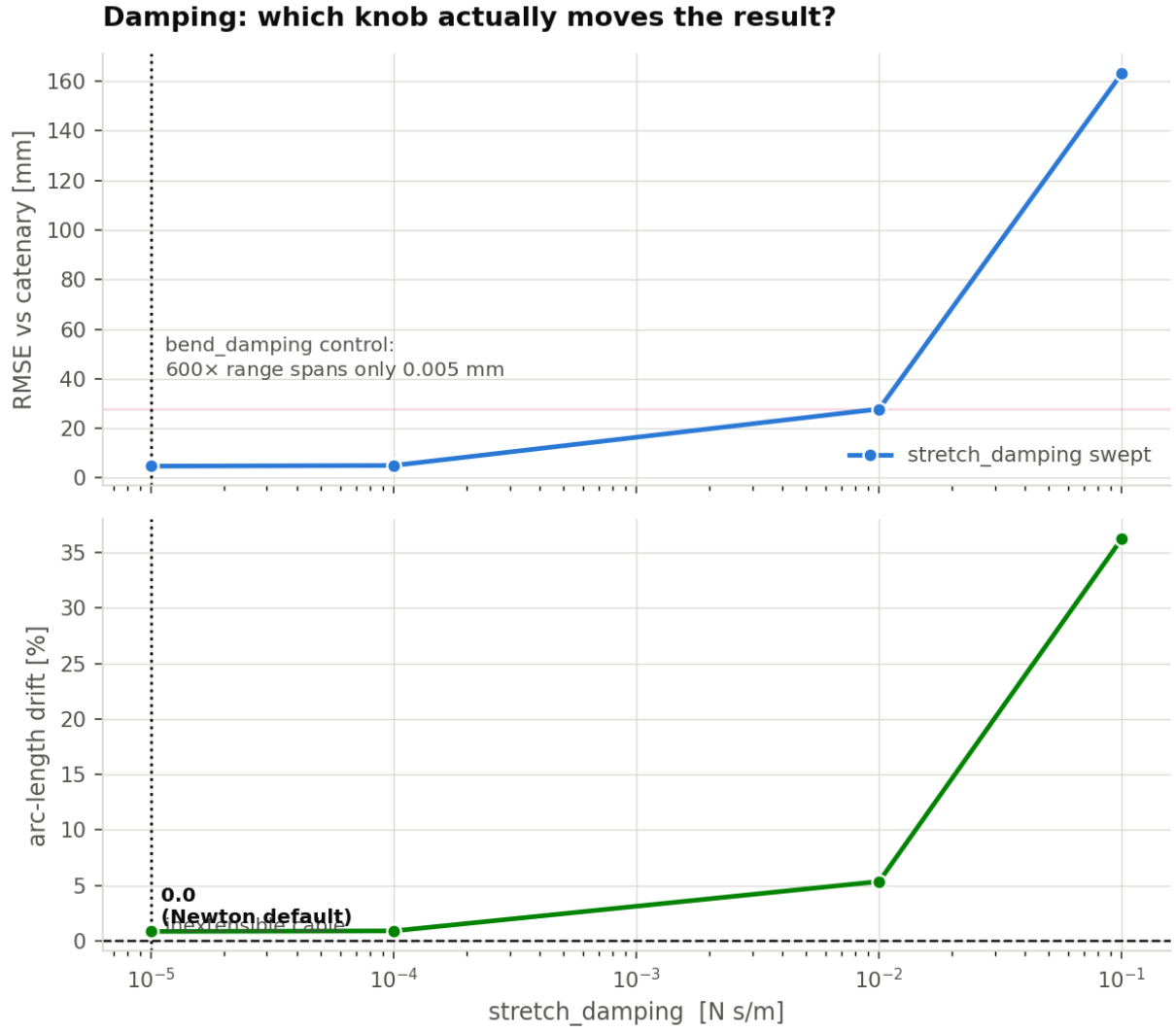


Figure 6: Damping. Stretch damping dominates; the bend-damping control spans a 600x range and does nothing.

field	source	value
cable.length_m	measured	0.8932
supports.mid_fraction	measured	0.4688

Table 4: Which inputs were measured from the photograph and which were assumed. Recorded automatically for every run.

here: the tape clamps the cable's *angle* as well as its position, so a slack cable leaves the clamp along the tape's direction and bends back, carrying it horizontally *past* its own anchor (86 px in this photograph). Taking the horizontal extreme therefore picks a point on the descending cable rather than the clamp.

An earlier version did exactly that at one end while the other came from where tracing halted — two ends defined by two different rules — and the resulting frame was tilted enough to place two physically level clamps 81 mm apart in height. With both ends found by one rule, and the clamp-to-clamp line adopted as the horizontal (the clamps are level in the apparatus, so this also removes camera roll, here -0.40°), that difference is now 0.0 mm by construction.

Independent check. The frame is fixed by the two OUTER clamps alone, so the middle clamp is free to disagree, and is therefore a genuine test. It is recovered at $x = 0.164$ m against 0.150 m measured by tape — 14 mm, or 4% of the span — and 3.9 mm above the outer clamps against 0 expected. Neither is forced by the calibration. The residual is consistent with hand-measuring the clamp positions and with lens distortion toward the frame edges, and it bounds the accuracy of any simulation-versus-photograph comparison below.

What the model-free check says. The traced arc length exceeds the length implied by fitting catenaries to the measured sags by roughly 58 mm. This is not numerical: it is the clamped end again. A catenary meets its support at whatever angle the length demands, while the tape forces a different angle and the cable bends to accommodate it, and that bend costs arc length the catenary does not spend. It is the largest single modelling discrepancy in this study and is a property of the apparatus, not of any solver.

Independent of the photograph entirely:

- the verification of (1) against its closed form;
- every solver's error against the analytic catenary;
- the hyperparameter sweep and its conclusions;
- the observation that the Warp rod diverges at this slackness, rising above the clamps rather than hanging between them.

Modelling limits. The catenary is the $EI \rightarrow 0$ limit. The real cable is a charger cable with a permanent set, so its rest shape is not straight and some deviation from any ideal curve is physical rather than numerical. The tape also clamps the cable's angle, which the classical catenary does not model. Both effects are concentrated near the clamps, and both are measurable once the frame is corrected — which makes them results to quantify rather than caveats to apologise for.